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Research and Applications

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SMART: a new patient similarity estimation framework for enhanced predictive modeling in acute kidney injury

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Abstract

Objective: Accurately measuring patient similarity is essential for precision medicine, enabling personalized predictive modeling, disease subtyping, and individualized treatment by identifying patients with similar characteristics to an index patient. This study aims to develop an electronic health record-based patient similarity estimation framework to enhance personalized predictive modeling for Acute Kidney Injury (AKI), a complex and life-threatening condition where accurate prediction is critical for timely intervention.

Materials and Methods: We introduce **S**imilarity **M**easurement for **A**cute Kidney Injury **R**isk **T**racking (SMART), a new patient similarity estimation framework with 3 key enhancements: (1) overlap weighting to adjust similarity scores; (2) distance measure optimization; and (3) feature type weight optimization. These enhancements were evaluated using internal and external validation datasets from 2 tertiary academic hospitals to predict AKI risk across varying group sizes of similar patients.

Results: The study analyzed data from 8637 patients in the reference patient pool and 8542 patients in each of the internal and external test sets. Each enhancement was independently evaluated while controlling for other variables to determine its impact on prediction performance. SMART consistently outperformed 3 baseline models on both the internal and external test sets ($P < .05$) and demonstrated improved performance in certain subpopulations with unique health profiles compared to a traditional machine learning approach.

Discussion: SMART improves the identification of high-quality similar patient groups, enhancing the accuracy of personalized AKI prediction across various group sizes. By accurately identifying clinically relevant similar patients, clinicians can tailor treatments more effectively, advancing personalized care.

Introduction

Electronic health record (EHR) systems have become a cornerstone to modern healthcare. With the rich, longitudinal, and multidimensional data, EHRs offer unparalleled insights into individual health trajectories, making them invaluable for disease risk and outcome prediction.[1–3] While advances in machine learning have improved the performance of prediction models, these models often target the average patient and may not perform equally well on individuals with unique health profiles.[4,5] Leveraging information from patients with similar characteristics ("patients like me") enables personalized medicine with individualized risk predictions and tailored treatment planning.[6–8]

Accurate estimation of patient similarity is critical for personalized medicine, as it directly determines the quality of the similar patient group used for decision-making.[9–11] Previous studies have developed personalized prediction models based on various patient similarity estimation methods. For instance, Ng et al. used metric learning and feature filtering to estimate patient similarity and construct a personalized logistic regression (LR) model for predicting diabetes onset.[12] David et al. employed the Euclidean distance on weighted features to select a group of most similar patients for an index patient.[13] Lee et al. developed a cosine-similarity-based personalized prediction model to forecast intensive care unit mortality.[14] Others have incorporated expert knowledge to weight features differently[15–17] or optimized the group size for similar patient estimation.[18] Wang et al. demonstrated that identifying a group of similar patients by optimizing the most predictive feature types and combining weights of these feature types achieved better predictive performance.[19]

However, none of these studies have considered the impact of missing data on patient similarity calculation. Missing data is unavoidable in real-world EHR and can significantly affect the estimation of patient similarity.[20–22] For example, 2 dissimilar patients may appear erroneously similar if missing data are replaced with imputed values based on population median or mean. Previous studies have explored ways to leverage the missingness in EHR data to enhance predictive modeling, such as incorporating "informative missingness" as a feature within prediction models.[23,24]

In this study, we developed **S**imilarity **M**easurement for **A**cute Kidney Injury **R**isk **T**racking (SMART), a new framework designed to optimize similar patient identification and

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enhance personalized prediction of Acute Kidney Injury (AKI), a complex and life-threatening syndrome.[25] SMART encompasses 3 key enhancements: (1) **o**verlap rate **w**eighting (**OW**) to adjust patient similarity scores. The overlap rates are calculated based on the patterns of missing data between pairs of patients. This approach is guided by the intuition that dissimilar patients are more likely to exhibit different patterns of missing values, as similar patients often undergo the same clinical protocols, resulting in similar data collection patterns;[26] (2) **d**istance measure **o**ptimization (**DO**) for selecting the best similarity calculation method; and (3) **f**eature type weights **o**ptimization (**FO**) to effectively combine feature types for different similar patient group sizes.

We evaluated SMART's ability to retrieve similar patients by framing it as a binary classification task, employing k-nearest neighbor (KNN) to simulate case-based reasoning by physicians and LR for personalized AKI onset prediction based on the identified similar patient groups.

To assess the impact of SMART's 3 key enhancements, we employed a "one-at-a-time" approach, assessing each enhancement independently while controlling for others. Additionally, we assessed SMART's effectiveness and robustness across 11 subpopulations, each characterized by distinct health profiles.

This study advances personalized AKI prediction by leveraging missing data patterns for similar patient identification, optimizing distance measures and feature weights to meet diverse clinical needs, and evaluating prediction performance across varied subpopulations. These contributions pave the way for leveraging similar patients and predictive analytics to improve AKI decision-making and outcomes.

Methods

Data source and processing

The patient cohort for this study comprised 3 parts: the patient pool (source for retrieving similar patients), the internal test dataset, and the external test dataset. Inpatient data from the University of Kansas Medical Center (KUMC) collected between January 1, 2015, and December 31, 2015, constituted the patient pool, while data from the same center collected from January 1, 2016, to December 31, 2016, formed the internal test dataset. The external dataset included inpatient data from the University of Pittsburgh Medical Center (UPMC), collected during the same period as the internal test set. Both the internal and external test datasets were of equal size. We used this temporal testing approach to simulate real-world scenarios in which only historical encounters can be used to match new encounters. Compared to the traditional random train-test splitting approach, this approach evaluates the framework's performance under potential data drift conditions. Furthermore, the external dataset provided an additional evaluation, testing the framework's robustness when applied to data with different patient distributions from another hospital.

AKI onset was defined using the SCr criteria of the Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guidelines.[27] For AKI patients, the prediction event time was the AKI onset day, while for non-AKI patients, it was the day of their last SCr measurement. This ensures fairness in the prediction event timing between AKI and non-AKI patients. The prediction point was set at 2-days before the prediction event to allow physicians sufficient lead time (ie, 1-day excluding the prediction day) to conduct risk assessment and design intervention. A 7-day observation window, moving backward from the prediction point, was defined to collect patients' daily mean SCr trajectories and most recent lab results as predictors, as these are the 2 most frequently used feature types for predicting AKI onset.[4,5,28,29] The inclusion criteria were (1) age over 18 years; (2) baseline SCr < 3.5 mg/dL; and (3) at least 2 days of SCr measurements within the 7-day observation window. To ensure patient uniqueness, only the first hospitalization encounter per patient was included. Lab tests with missing rates exceeding 50% were discarded to minimize the influence of imputed values on patient similarity estimation.[30] Various forms of SCr and glomerular filtration rate (GFR) measurements were excluded to avoid redundancy, as SCr trajectories were already considered as an independent feature. The final lab test feature space defined in the patient pool was consistently applied across the internal and external test datasets to ensure uniformity.

Overview of the framework

The previous patient similarity framework proposed by Wang et al. only considered the optimal weights for combining different feature types at a fixed group size k.[19] However, the group size varies in reality depending on the complexity of the patient's case. In this study, we introduced 3 key enhancements to improve the quality of similar patient groups and enhance AKI prediction across a wide range of k: (1) OW to adjust patient similarity scores, (2) DO to ensure that the optimal distance measure is selected for calculating patient similarity; and (3) FO to effectively combine feature types according to different group sizes. The workflow of SMART is illustrated in Figure 1.

Overlap rate calculation (OW)

SCr overlap rates

We first calculated the data overlap rates for the 2 feature types separately to adjust patient similarity scores, the calculation of which will be described in the Distance measures and similarity scores section. For simplicity, we will illustrate this process using the i-th and j-th patients. Let $\mathrm{SCr}_i = \left[s_{-8}^{(i)}, s_{-7}^{(i)}, \ldots, s_{-2}^{(i)}\right]$, where $s_t^{(i)} \in \{0,1\}$, denote the binary vector indicating the days when SCr measurements are available for the i-th patient. Here, the subscript t (ranging from −8 to −2) represents the day of the SCr measurement relative to the prediction event (ie, AKI onset or the last SCr measurement). We define $O_{\mathrm{SCr}}^{(i,j)} = \mathrm{SCr}_i \;\; \mathrm{SCr}_j$ as the overlapping positions in their SCr trajectories, where   denotes the logical "AND" operator. Many studies indicate that for in-hospital acute syndromes like AKI, changes in major biomarkers (ie, SCr in this case) within 48 hours before onset are more clinically meaningful and predictive.[27,31] Consequently, the impact of missing SCr measurements at different time intervals on the prediction of AKI varies, with greater impact the closer it is to the prediction point. To address this, we applied a Gaussian weighting function to the data overlap vector between 2 patients, where the weighting decreases as the time moves further away from the prediction point (Figure 2A). Specifically, if $O_{\mathrm{SCr}}^{(i,j)}$ contains K intervals of contiguous ones, the start and end indices of the k-th interval can be denoted by $a_k^{(i,j)}$ and $b_k^{(i,j)}$, respectively, with $a_k^{(i,j)}, b_k^{(i,j)} \in \{-8, \ldots, -2\}$, for $k = 1, \ldots, K$. The SCr trajectory overlap rate between these 2 patients is calculated as:

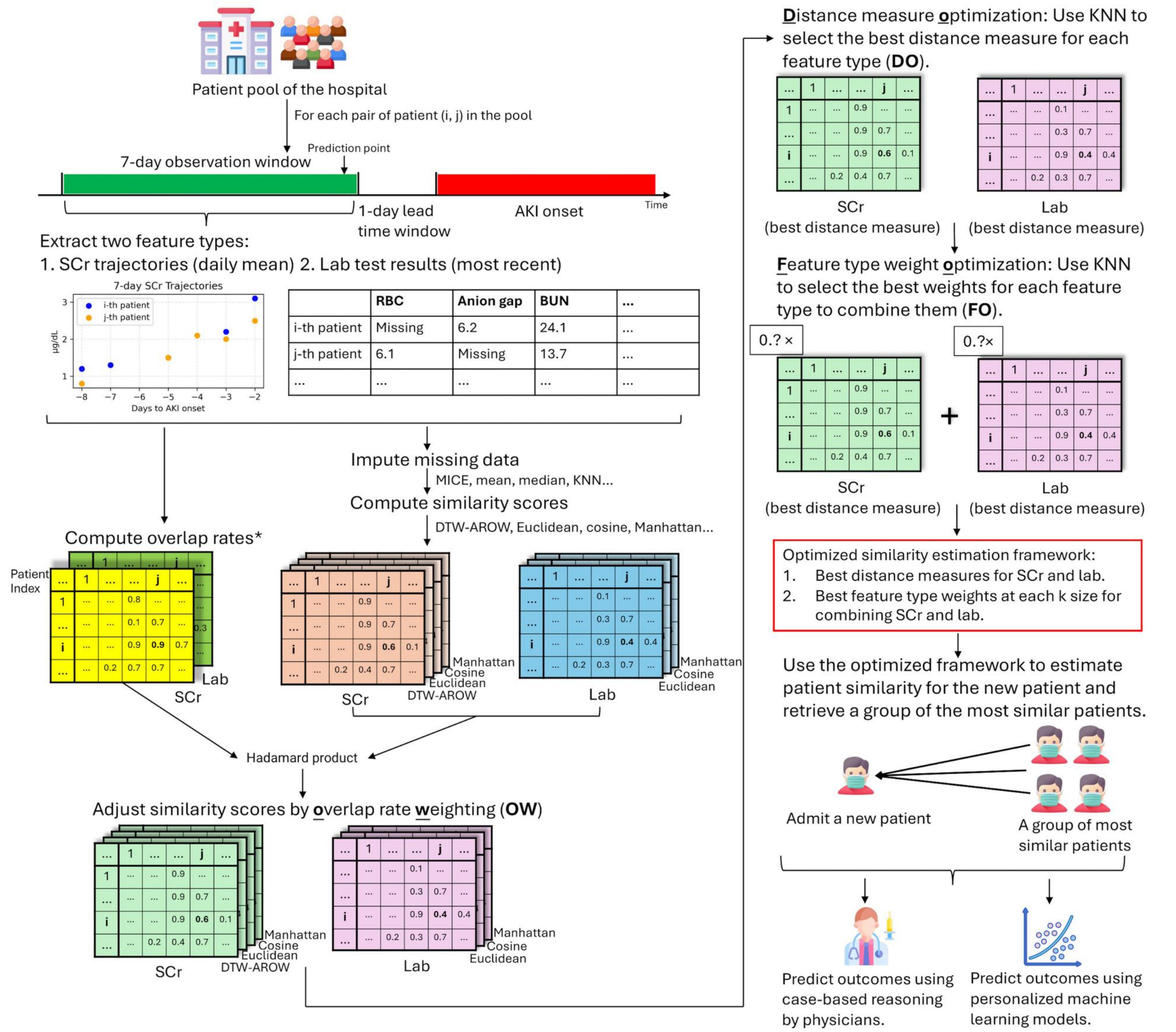

Figure 1. Overview of the SMART framework. *The procedures for computing overlap rates are visualized in Figure 2.

$$r_{\mathrm{SCr}}^{(i,j)} = \sum_{k=1}^{K}\left[\Phi\left(\frac{-a_k^{(i,j)}-2}{\sigma}\right)-\Phi\left(\frac{-b_k^{(i,j)}-2}{\sigma}\right)\right]$$

Here, $r_{\mathrm{SCr}}^{(i,j)}$ denotes the SCr trajectory overlap rate, Φ denotes the cumulative distribution function of the standard normal distribution, and σ is a scaling factor. For the following experiments and the final model, we set $\sigma = 1.5$. Additionally, we evaluated the impact of different σ values ($\sigma \in [1.2, 1.5, 1.7, 2.0, 2.5]$) on the model performance, as variations in σ significantly affect the contribution of overlapping SCr measurements from the first few days of the observation window to the computation of SCr overlap rates.

Lab overlap rates

For lab data, we used the same method as for SCr trajectories to derive the lab overlap vector, $O_{lab}^{(i,j)}$, between the i-th and j-th patients. Specifically, $O_{\mathrm{lab}}^{(i,j)} = \mathrm{Lab}_i \ \mathrm{Lab}_j$, where Lab_i and Lab_j are 2 binary vectors representing the availability of lab test results for the 2 patients. Since not all lab features have the same predictive power for AKI, we applied a random forest model on the patient pool to learn a set of feature weights, then selected the top 50% most important features, normalized their feature weights to a range between 0 and 1, and set the feature weights of the bottom 50% to 0 to obtain a feature weighting vector (Figure 2B). This method penalizes missing data only on the top 50% most important lab tests, with higher penalties for more important lab tests. Additionally, we examined the impact of varying this threshold (top 10%, 30%, 50%, 70%, and 100%) on the model's performance. This feature weighting method is flexible and can be replaced by other approaches, such as incorporating expert knowledge. The final lab overlap rates are derived as follows:

$$r_{\mathrm{lab}}^{(i,j)} = O_{\mathrm{lab}}^{(i,j)} \cdot W_{\mathrm{lab}}$$

Here, $r_{\mathrm{lab}}^{(i,j)}$ denotes the lab overlap rate, and W_{lab} denotes the normalized feature weighting vector.

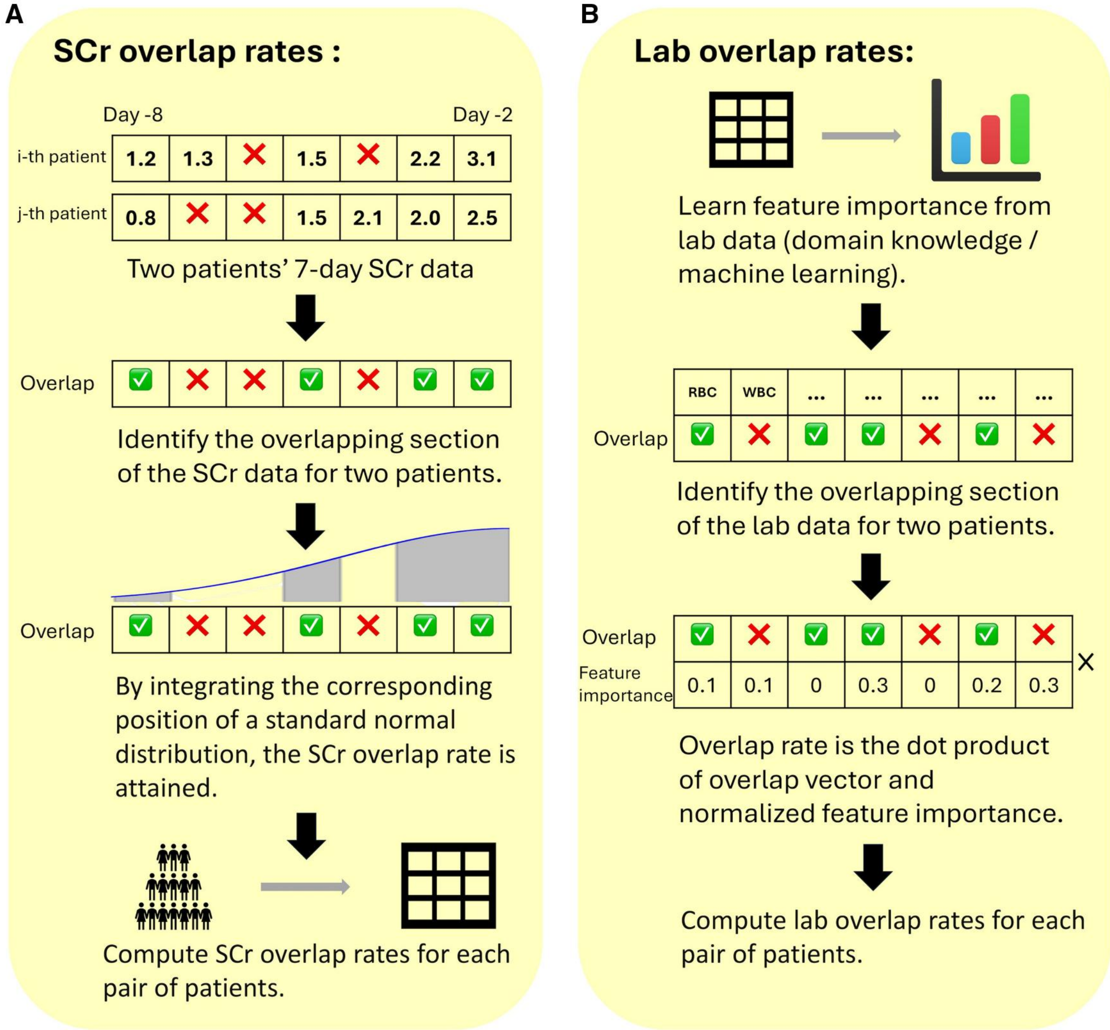

Figure 2. Procedures for calculating overlap rates for 2 features.

Patient similarity calculation

Missing data imputation

Before calculating the similarity scores between 2 patients, we performed data imputation for missing values, as some distance measures, such as Euclidean and cosine distance, do not support calculations with missing data. For SCr trajectories, we used a combination of linear interpolation, last observation carried forward, and first observation carried backward to impute missing data at different positions.[32] A copy of the raw, unimputed SCr trajectories was retained for use in the Dynamic Time Warping with Additional Restrictions on Warping (DTW-AROW) method, as described in the following section.[33] For lab features, we used the Multivariate Imputation by Chained Equations.[34] We also evaluated whether other missing data imputation methods (median, mean and KNN imputation for both SCr and lab features) significantly affected the performance of our framework.

Distance measures and similarity scores

In this section, we introduce the distance measures and procedures for calculating similarity scores for the 2 feature types. These similarity scores were used in the DO and FO steps, which are detailed in the next section. We evaluated 3 widely applied distance measures for both feature types: Euclidean distance, cosine distance, and Manhattan distance. Additionally, since SCr trajectories are time-series data, we also applied the DTW-AROW method on the raw, unimputed SCr trajectories. DTW-AROW is an extension of traditional Dynamic Time Warping (DTW),[35] a technique that aligns time series by minimizing the distance between them through non-linear temporal mapping. It is particularly well-suited for handling variable-length time series resulting from missing data. The descriptions of these distance measures are provided in Supplementary Method 1.

To ensure that each lab feature contributed equally to the distance calculation, the lab features were min-max normalized prior to the distance calculation. The distance between the *i*-th patient and the *j*-th patient, calculated using a specific distance measure for each feature type, was normalized to a range between 0 and 1 and then subtracted from 1 to obtain the similarity score between them. This similarity score was further adjusted by the corresponding SCr, or lab overlap rate. The process is described as follows and is applied to both feature types:

$$d'_{i,j} = \frac{d_{i,j} - \min_c d_{i,c}}{\max_c d_{i,c} - \min_c d_{i,c}}$$

$$s_{i,j} = 1 - d'_{i,j}$$

$$s'_{i,j} = r_{i,j} \cdot s_{i,j}$$

Here, $d_{i,j}$ denotes the distance for either of the 2 feature types, calculated using a specific distance measure, c denotes the patient index, and $d'_{i,j}$ denotes the normalized distance. $s_{i,j}$ denotes the similarity score. $r_{i,j} = r^{(i,j)}_{\mathrm{SCr}}$ or $r^{(i,j)}_{\mathrm{lab}}$, representing the overlap rate for the corresponding feature type, and $s'_{i,j}$ denotes the OW-adjusted similarity score for either of the 2 feature types.

Distance measure and feature weight optimization (DO and FO)

In this step, we optimized the distance measures for each feature type and the weights used to combine them. We first optimized the distance measures for each feature type, using the KNN model as the base model. We evaluated Euclidean distance, cosine distance, Manhattan distance for both SCr trajectories and lab tests, as well as DTW-AROW for SCr trajectories only, across k values of 10, 20, …, 200. The evaluation focused on predicting AKI onset for the i-th patient using each feature type independently, ie, based on the top k rankings of either $s'_{(i,c),\ \mathrm{SCr}}$ or $s'_{(i,c),\ \mathrm{lab}}$ (where c index all other patients), to retrieve k similar patients and make predictions. Based on the performance of different distance measures for each feature type, we selected the optimal distance measure at a specific k value for that feature type. The distance measure that performed best across the majority of k values was chosen as the overall optimal distance measure for that feature type.

Next, we optimized the feature type weights to combine the similarity scores calculated using the optimal distance measure to derive the final patient similarity score between the i-th and j-th patients, $s^{\mathrm{final}}_{i,j}$, as described below:

$$s^{\mathrm{final}}_{i,j} = \lambda_{SCr} \cdot s'_{(i,j),\ \mathrm{SCr}} + \lambda_{\mathrm{lab}} \cdot s'_{(i,j),\mathrm{lab}}$$

Here, λ_{SCr} and λ_{lab} denote the weights for the corresponding feature types. The k most similar patients to an index patient were identified based on the rankings of the final similarity scores. We evaluated 9 sets of different weight combinations: ($\lambda_{\mathrm{SCr}} = 0.1,\ \lambda_{\mathrm{lab}} = 0.9$), ($\lambda_{\mathrm{SCr}} = 0.2,\ \lambda_{\mathrm{lab}} = 0.8$), …, ($\lambda_{\mathrm{SCr}} = 0.9,\ \lambda_{\mathrm{lab}} = 0.1$). Unlike distance measures, where a single overall optimal measure was applied across all group sizes k, the feature type weights were optimized separately for each investigated k value. As a result, each k value had its own unique set of optimal weights.

For both optimization tasks (DO and FO), we used the Area Under the Precision-Recall Curve (AUPRC) as the optimization objective, as it emphasizes the prediction performance for the positive class (ie, patients with AKI) in imbalanced data and evaluates the overall model performance across different thresholds. Both the best distance measure for each feature type and the optimal feature type weights for each k value were determined using the leave-one-out method on the patient pool.

Performance evaluation

We evaluated the performance of our framework on both internal and external test datasets. After DO and FO, we have identified the optimal distance measure for each feature type and the optimal feature type weights across different k sizes. Using these parameters, each patient in the test sets selected $k = 10$, 20, …, 200 most similar patients from the patient pool to predict the risk of AKI using either KNN-voting (simulating case-based reasoning by physicians) or a LR model (for personalized predictive modeling). Both AUPRC and the area under the receiver operating characteristic (AUROC) were used as performance metrics.

First, to assess whether the 3 key enhancements (ie, OW, DO, and FO) could individually improve model performance, we evaluated the impact of each enhancement while controlling for other variables. Then, to determine whether the final SMART framework incorporating all 3 key enhancements demonstrates superiority over other personalized modeling approaches, we selected 3 models as baseline comparisons: the Euclidean distance-based personalized LR,[13] the cosine distance-based personalized LR,[14] and a personalized LR model that used predefined distance measures (DTW-AROW for SCr and Euclidean distance for lab) and optimized feature type weights at a single k value ($k = 20$), which were then applied to other k sizes.[19] The distance calculation for the first 2 baseline models was performed in the normalized combined SCr-lab feature space, whereas the third baseline model treated SCr and lab as 2 separate feature types, in the same way as SMART. Additionally, SMART was compared against 2 ablation study models: the DO-only and the DO+FO models. The DO-only model fixed its feature type weights to the optimal weights at $k = 20$ across all k sizes, the same as the third baseline model, but still incorporating the DO mechanism. We conducted a one-tailed paired t-test to determine whether the observed performance enhancements were statistically significant compared to the baseline and ablation study models, with a significance level of $P < .05$ set as the threshold for statistical significance.

Furthermore, we analyzed the framework's performance across 11 subpopulations and compared it to a traditional global LR trained on 100% of the training data to evaluate whether SMART offers an advantage over the traditional "one-size-fits-all" approach for specific subpopulations with unique health profiles. The performance was assessed at both $k = 50$ and $k = 100$. These subpopulations included male patients (as females are believed to be protected from AKI),[36] black patients, patients aged over 70 years, and patients with cancers, chronic kidney diseases (CKD), chronic liver diseases, circulatory system diseases, congestive heart failure, diabetes mellitus, hypertensive diseases, and metabolic diseases. The codes used to define these subpopulations are provided in Table S1.

Results

Study cohorts

The patient pool comprised a total of 8637 patients with at least one hospitalization record at KUMC between January 1, 2015, and December 31, 2015. The internal test set included 8542 patients hospitalized at KUMC between January 1, 2016, and December 31, 2016. The external test set consisted of 8542 patients hospitalized at UPMC during the same period as the internal test set. The proportions of

Table 1. Demographic information.

	Patient pool	Internal test set	External test set	*P*-values
Source	KUMC	KUMC	UPMC	/
Cohort size	8637	8542	8542	/
Time window	January 1, 2015-December 31, 2015	January 1, 2016-December 31, 2016	January 1, 2016-December 31, 2016	/
Number of lab features	40	40	40	/
SCr trajectory days	7	7	7	/
Averaged SCr levels 48h prior to the prediction points (mg/dL)	0.87 (0.36)	0.86 (0.35)	0.94 (0.42)	/
AKI rates (%)	8.07	7.98	7.86	NS
Male (%)	51.33	51.88	48.28	<.001
Black race (%)	13.33	12.71	9.55	<.001
Age ≥ 70 (%)	28.27	29.43	45.15	<.001
Cancers (%)	31.39	29.75	32.03	<.01
CKDs (%)	3.02	3.24	6.39	<.001
Chronic liver diseases (%)	4.77	6.51	4.43	<.001
Circulatory system diseases (%)	10.39	9.79	17.54	<.001
Congestive heart failure (%)	3.44	4.59	2.45	<.001
Diabetes mellitus (%)	10.39	9.85	15.37	<.001
Hypertensive diseases (%)	25.43	24.83	42.59	<.001
Metabolic diseases (%)	19.81	18.65	33.51	<.001
Peripheral vascular diseases (%)	1.83	1.35	4.65	<.001
Stroke (%)	1.68	1.29	1.10	<.01

The numbers in parentheses represent the SD. *P*-values were derived from the chi-square test.
Abbreviation: NS, not significant.

patients who experienced AKI during hospitalization were comparable: 8.07% in the patient pool, 7.98% in the internal test set, and 7.86% in the external test set. The patient pool and internal test set had a significantly higher proportion of patients identifying as Black ($P < .001$), while the external test set had a significantly higher proportion of patients aged over 70 years ($P < .001$). Details of the patient cohort are provided in Table 1. Data missingness varied across the 3 datasets (Figure S1), with the external test set exhibiting higher levels of missing data compared to the patient pool and internal test set ($P < .001$).

OW mechanism improves the personalized model performance

SCr overlap rate weighting

The distributions of SCr overlap rates are presented in Figures S2A-C. The median SCr overlap rates across the 3 cohort pairs (patient pool-patient pool, patient pool-internal test set, patient pool-external test set) were ∼0.8.

Using the internal test set, we evaluated the impact of OW across different distance measures including DTW-AROW, Euclidean distance, cosine distance, and Manhattan distance. LR was applied as the final model, and AUPRC was used as the evaluation metric. The OW scenario significantly outperformed the no-OW scenario across all tested standard normal distribution scaling factor (σ) settings, except for cosine distance (Figure 3A, B, and D, all $P < .001$). The largest improvement was observed with DTW-AROW, showing an average AUPRC increase of 26% across all tested k sizes when $\sigma = 1.5$. For cosine distance (Figure 3C), significant improvements under the OW scenario were observed only at $\sigma = 1.2$ ($P < .001$), $\sigma = 1.5$ ($P < .001$), and $\sigma = 1.7$ ($P < .05$). Varying σ did not significantly impact performance across any of the 4 distance measures on SCr overlap rates.

Sensitivity analyses of the OW mechanism on SCr using KNN-voting as the final model, and LR evaluated by AUROC are shown in Figures S3-S5. Results are generally consistent with those observed when AUPRC was used as the metric and LR as the final model.

The effects of 3 alternative missing data imputation methods (ie, median, mean, and KNN imputation) on OW were evaluated using $\sigma = 1.5$ (Figures S6-S17). With median imputation, OW demonstrated significant performance improvements across all conditions, except when Manhattan distance was used and KNN-voting was the final model (Figures S6-S9, all $P < .001$). Similar trends were observed with mean imputation (Figures S10-S13). With KNN imputation, OW enhanced model performance exclusively with the DTW-AROW distance measure (Figures S14-S17, $P < .001$).

On the external test set, OW showed mixed results depending on the final model used (Figures S18-S21). When employing KNN-voting, OW failed to outperform the no-OW scenario when using Euclidean and Manhattan distances (Figures S18 and S19, NS). In contrast, when LR was used as the final model, significant performance improvements were observed across all 4 distance measures (Figures S20 and S21, $P < .05$).

Lab overlap rate weighting

The distributions of lab overlap rates are presented in Figures S2D-F. The lab overlap rates were significantly lower in the patient pool-external test set pair compared to the patient pool–patient pool and patient pool–internal test set pairs (both $P < .001$).

On the internal test set, OW consistently outperformed the no-OW scenario across all tested threshold settings (Figure 3E-G). Specifically, computing lab overlap rates using a feature importance threshold of 100% (ie, all features included) yielded the best performance (all $P < .001$). Sensitivity analyses using AUROC as the metric and KNN-voting as the final model further supported these findings Figures S22-S24. Significant improvements were also observed across all 3 missing data imputation methods when using a threshold of 50% (Figures S25-S36).

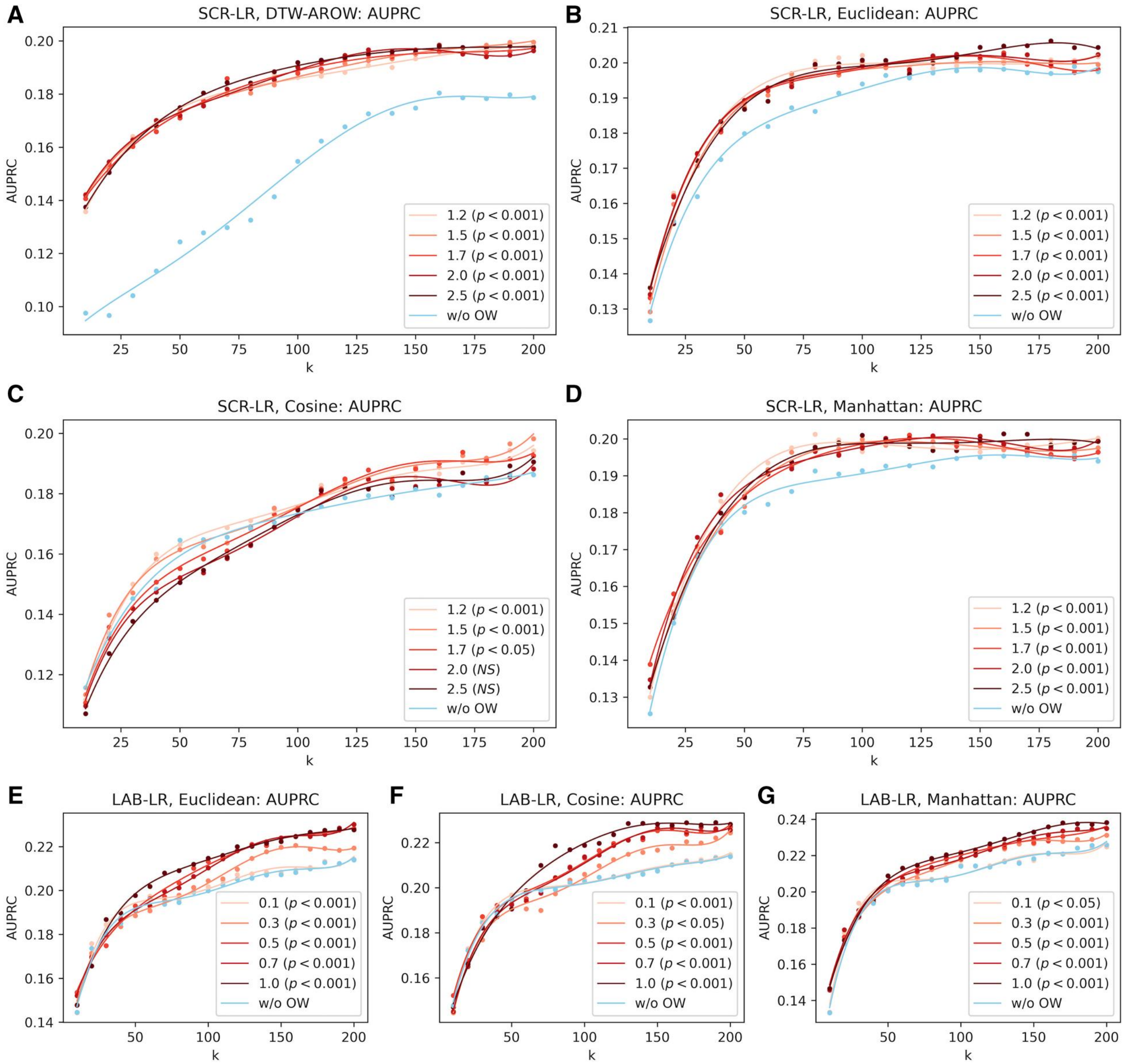

Figure 3. Impact of 2 hyperparameters for computing overlap rates on model performance. The base model is LR, and the evaluation metric is AUPRC. For SCr (**A-D**), the hyperparameter is the scaling factor (σ) of the standard normal distribution. With $\sigma \in [1.2, 1.5, 1.7, 2.0, 2.5]$, the last 2 days of the observation window' s SCr measurements (days -3 and -2) accumulate overlap rates of 0.90, 0.82, 0.76, 0.68, and 0.58, respectively. For lab (**E-G**), the hyperparameter is the percentage of the most important features to be considered. The *P*-values indicate whether the current hyperparameter setting achieved significantly better performance compared to the no-OW scenario.

On the external test set, significant performance gains were achieved using KNN-voting with Euclidean ($P < .05$) and Manhattan distances ($P < .001$) and using LR with Manhattan distance ($P < .001$). However, in other cases, OW did not significantly outperform the no-OW scenario (Figures S37-S40).

Distance optimization and FO improve personalized model performance

On the internal test set, we observed that the DO mechanism did not always select the best-performing distance measure (Figures S41-S44) but effectively eliminated poorly performing ones. For instance, while DTW-AROW did not consistently outperform other measures across all *k* sizes, the DO mechanism excluded the worst-performing cosine distance (Figures S41c,d). The number of times different distance measures were selected as the best measure across all *k* sizes is shown in Figures S45. Similarly, the FO mechanism successfully excluded suboptimal feature weight combinations, even if it did not always yield the best-performing feature weights on the test set (Figure S46). With increasing *k* sizes, lab features received greater weight in the FO process (Figure S47).

Final framework performance analysis

The SMART framework (DO+FO+OW) demonstrated significantly superior performance on the internal test set (Figure 4) compared to all 3 baseline models (base model, Euclidean-distance model, and cosine-distance model; all $P < .001$) and the 2 SMART ablation study models (DO and DO+FO; both $P < .001$). While the DO mechanism alone provided substantial improvement over the base model, which utilized the predefined distance measures DTW-AROW for SCr and Euclidean distance for lab, the FO

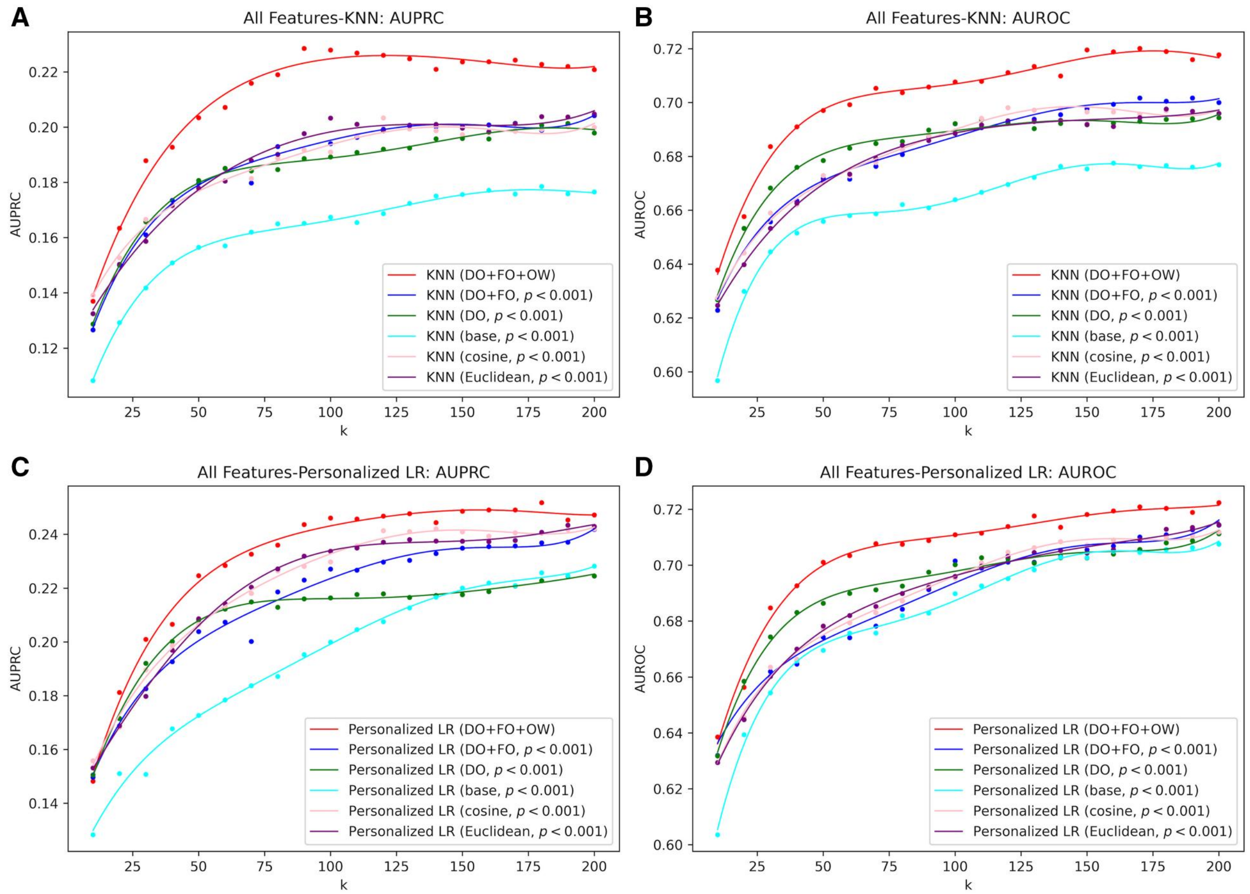

Figure 4. Final model performance on the **internal** test set. The *P*-values indicate whether the final SMART framework (DO+FO+OW) achieved significantly better performance compared to the corresponding baseline or ablation study models. **(A, B)** KNN. **(C, D)** Personalized LR.

mechanism did not consistently outperform the DO-only model. The traditional Euclidean- and cosine-distance-based personalized models, which calculated patient distance in the combined SCr-lab feature space, outperformed the SMART ablation study models but were still surpassed by the full SMART model.

On the external test set (Figure 5), SMART outperformed all other models when AUPRC was used as the evaluation metric (all $P < .001$). However, when evaluated using AUROC, it achieved performance comparable to the DO-only model (NS). Models that treated SCr and lab as separate feature types (base, DO, and DO+FO+OW) significantly outperformed traditional personalized models based on applying Euclidean or cosine distances in the combined SCr-lab feature space. However, the FO mechanism when applying optimal feature type weights derived from a different hospital resulted in decreased model performance.

Final framework performance in subpopulations

Using a group size $k = 100$, AUPRC as the evaluation metric and KNN-voting as the final model, SMART outperformed all baseline models across all subpopulations (Figure 6A). When evaluated using AUROC, improvement was observed across most subpopulations except for Black patients, diabetes mellitus patients, and chronic liver disease patients (Figure 6B). Similar trends were observed when LR was used as the final model (Figures 6C and D), as well as when using a group size $k = 50$ (Figure S48a,b).

We also compared SMART with the traditional global LR model trained on 100% of the training data and evaluated on the test set (green lines in Figures 6 and Figure S48c-f). The SMART model, utilizing a personalized approach, demonstrated advantages over the global LR model in 4 disease-specific subpopulations based on both AUPRC and AUROC: patients with congestive heart failure (5%), circulatory system diseases (10%), metabolic diseases (19%), and hypertensive diseases (25%). However, the global LR model outperformed SMART in certain broader subpopulations, including Black patients (13%), patients aged 70 or older (29%), cancer patients (30%), and male patients (52%).

Discussion

In this study, we proposed SMART, a framework designed to identify a group of similar patients for AKI risk prediction of an index patient and provide personalized decision-making support. SMART incorporates 3 key enhancements—DO, FO, and OW—targeting 2 main objectives: improving model accuracy and addressing varying clinical needs as reflected by the size of the similar patient group to retrieve. Compared to the baseline models, SMART demonstrated superior performance not only in the general population but also in specific subpopulations with unique health profiles, highlighting its adaptability and effectiveness across diverse clinical contexts. We discuss key findings from our experiments below.

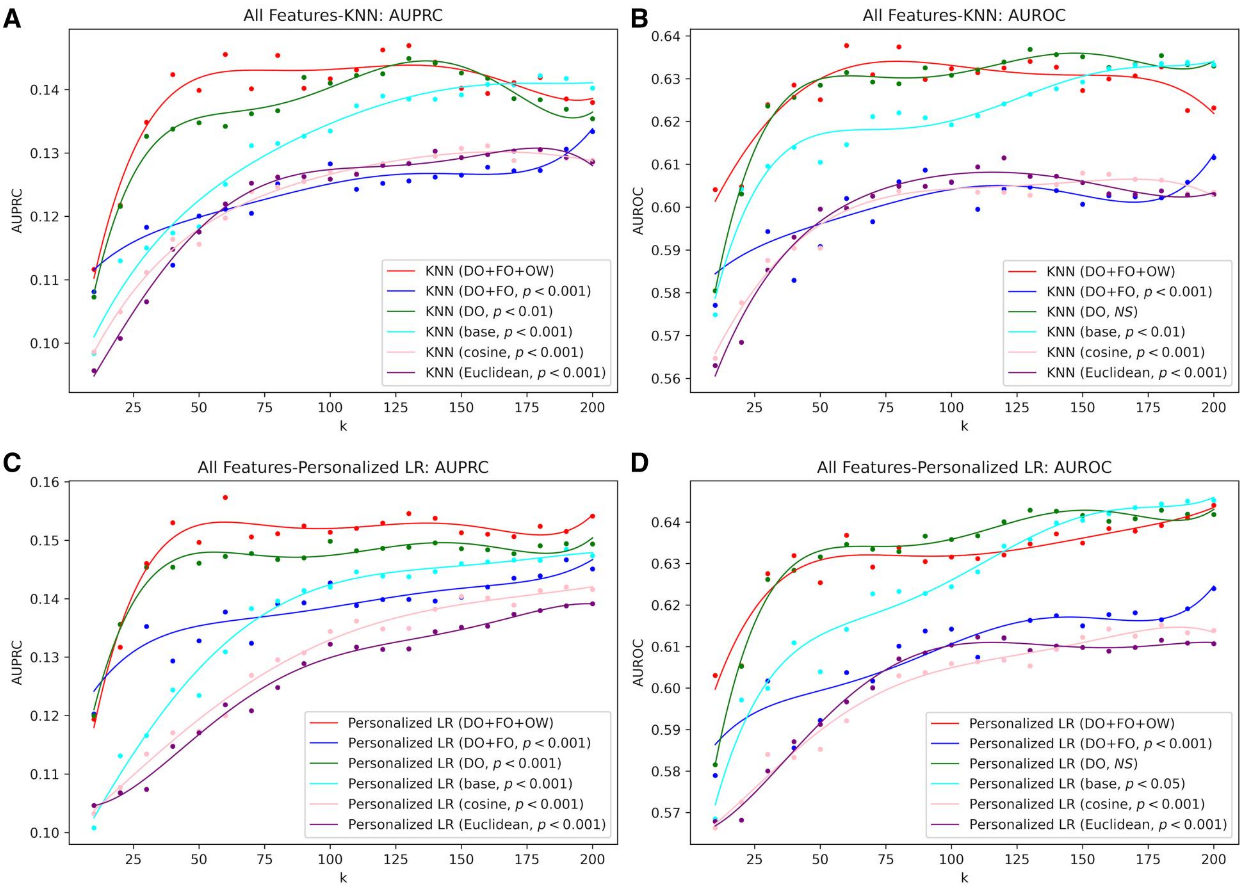

Figure 5. Final model performance on the **external** test set. The *P*-values indicate whether the final SMART framework (DO+FO+OW) achieved significantly better performance compared to the corresponding baseline or ablation study models. (**A, B)** KNN. (**C, D)** Personalized LR.

The variation in σ did not significantly affect the OW mechanism's performance on SCr (Figure 3 and Figures S3-S5). On the patient pool and internal test set, fewer than 10% of patients had missing SCr for the last 2 days of the 7-day observation window (Figure S1a,b). Consequently, even at $\sigma = 2.5$, the penalty for collectively missing the remaining 5 days only accumulated to 0.42. Additionally, since the analysis considered up to 200 most similar patients, the OW mechanism primarily adjusted for dissimilar patients that were erroneously included in the top k patients due to missing data imputation. This suggest that most of the similar patient group remained genuinely similar, limiting the impact of varying σ on model performance.

Computing lab overlap rates using all features (a threshold of 100%) consistently yielded the best performance compared to lower thresholds because the inclusion of all feature importance ensured a more comprehensive assessment between patients (Figure 3 and Figures S22-S24). Furthermore, using the full lab feature space allows better discrimination between patients with subtle differences in lower ranked features, enhancing stratification power and granularity.

The DO and FO mechanisms did not consistently select the best-performing distance measure for the test set, likely due to data drift. Since the patient pool and test datasets were from different years, variations in the underlying data characteristics could affect model performance.[37,38] This data drift can lead to changes in feature distributions and relationships over time. As a result, distance measures and feature-type weights optimized for the patient pool may not generalize effectively to the test set from a different year. This issue underscores the challenges posed by temporal variations in data distributions, which can affect the reliability and consistency of model performance across different datasets.

Heterogeneity in data distribution between hospitals can also lead to declines in model performance,[39] as observed in the external test set (UPMC), where the use of feature type weights derived from the KUMC patient pool resulted in reduced performance (Figure 5, the DO+FO model). This performance degradation is likely due to differences in data collection methods, lab testing protocols, and clinical practices between hospitals, leading to discrepancies in feature distributions and their relative importance.

A transfer learning approach can be explored in future research to alleviate this issue.[4,40] Within the SMART framework, which aims to optimize both distance measures and feature type weights for each hospital, 2 levels of knowledge transfer can be leveraged from a source hospital to a target hospital to mitigate performance degradation due to inter-institutional heterogeneity. Here, we define the target hospital as the one providing the patient pool and the source hospital as the origin of the external patient. First, the optimized distance measure and feature-type weights obtained at the source hospital can be transferred to the target hospital and integrated with the target-specific configuration. This may help bridge differences in optimal configurations arising

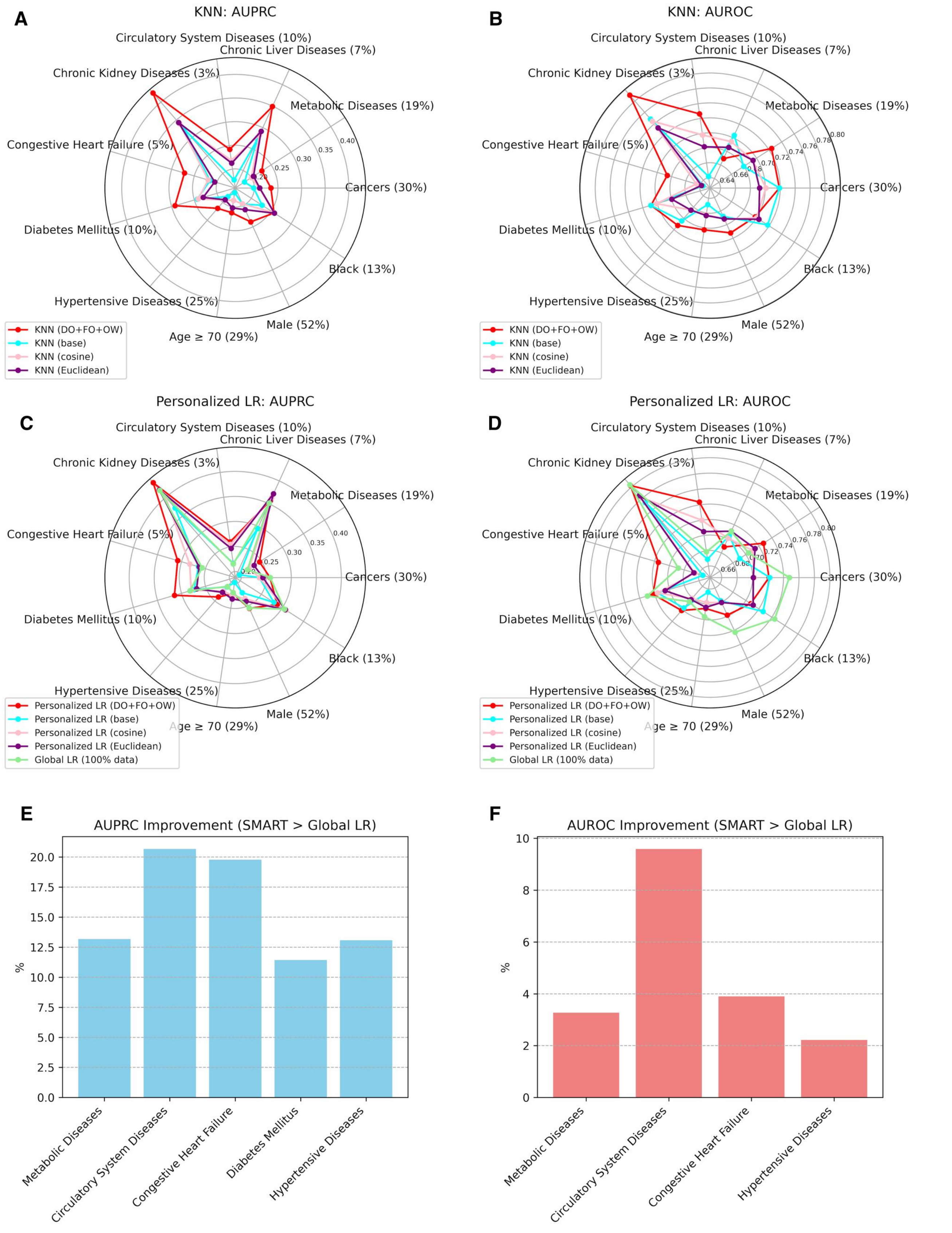

Figure 6. Final model performance in subpopulations with $k = 100$. The numbers in parentheses represent the percentage of the subpopulations. (**A, B**) KNN. (**C, D**) Personalized LR. (**E, F**) Bar charts display the subpopulations in (**C, D**) where SMART (Personalized LR with DO + FO + OW) outperforms the global LR. For AUPRC, only improvements exceeding 5% are shown.

from variations in data distribution across institutions. Second, if LR is used as the base model, the model parameters trained on the source hospital population can serve as initialization for the target hospital model. Fine-tuning can then be performed using the retrieved similar patient group. The rationale is that these parameters encode general population patterns from the source hospital, which can enhance model performance in the target setting by providing a more informed starting point for adaptation.

In the subpopulation analysis, we observed that SMART underperforms compared to the traditional global model in certain broader subgroups, such as Black patients (13%) and patients aged 70 or older (29%), as shown in Figure 6. A potential explanation is that these subpopulations comprise a substantial portion of the overall cohort and therefore exhibit data distributions that resemble the general population. They may benefit more from parameter estimation learned across the full population rather than from localized optimization. For such groups, personalized models like SMART—which rely on tailoring predictions based on a small cohort of similar patients—may offer limited added value, as the global model already captures representative patterns. Future research may focus on developing ensemble strategies that combine global and personalized models to adaptively balance between population-wide patterns and individual-level customization.

Although SMART demonstrated improved performance in both the general population and specific subpopulations, several limitations should be acknowledged: (1) Increased computational cost: Traditional personalized modeling incurs an approximate computational cost of $O(N)$, where N is the number of patients in the patient pool. In contrast, SMART's cost increases linearly with the number of feature types, M, as it processes each feature type individually before aggregating the results. Specifically, during the optimization phase, the computational complexity is $O(MN^2)$, primarily due to the leave-one-out strategy and the per-feature-type optimization. In the inference phase, the cost for a single patient is reduced to $O(MN)$, as the optimal configuration has already been determined. The complexity of SMART may result in high computational costs when the patient pool is large (ie, when N is large), and its translation to other diseases may be limited if multiple cross-sectional features are essential (ie, when M is large). Sampling a fraction of the patient pool during the optimization phase may help reduce computational costs. In the inference phase, the $O(MN)$ complexity remains sufficiently efficient to support practical deployment in clinical settings, particularly when M is not large, as it does not substantially exceed the complexity of traditional methods with $O(N)$ cost; (2) Sensitivity to data drift and heterogeneous data distributions: SMART may be affected by data drift and variations in data distributions across hospitals, potentially impacting its generalizability. However, these challenges are not unique to SMART and are also common to traditional personalized modeling approaches. The 2-level transfer learning approach proposed above may serve as an effective strategy to address this limitation.

Overall, SMART provides a strong foundation for future advancements in personalized prediction of AKI. By effectively integrating feature-specific insights and addressing challenges such as missing data in EHRs, SMART offers a promising framework for more accurate and individualized AKI risk prediction.

Author contributions

Mei Liu obtained funding for the project. Mei Liu and Deyi Li designed the overall study. Deyi Li performed data extraction, processing and all experiments. Alan Yu and Dana Fuhrman contributed their clinical expertise in analyzing results. Deyi Li drafted the manuscript with critical edits by Mei Liu, Alan Yu, and Dana Fuhrman.

Deyi Li (Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing—original draft), Alan Yu (Resources, Supervision, Validation, Writing—review & editing), and Dana Fuhrman (Resources, Supervision, Validation, Writing—review & editing), Mei Liu (Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing—review & editing)

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

A.Y. provides consultancy for Regulus Therapeutics Inc., Calico Life Sciences LLC, and Johnson & Johnson. These roles are not related to any material in this article. D.L., D.F., and M.L. declare no conflicts of interest.

Data availability

The data used in this study were de-identified EHRs from the Greater Plains Collaborative (GPC) and PaTH clinical research networks, that is not publicly available due to institutional and regulatory restrictions. Access to the data is governed by GPC and PaTH coordinating centers and requires institutional data use agreements and approval to protect patient privacy and compliance with ethical and legal considerations. Researchers interested in accessing similar data may contact GPC and PaTH front doors to inquire about potential data access. Source code for this study is available at https://github.com/GatorAIM/AKI_SMART.git.

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